

# Problem Set 5 – Proof and disproof

Session 5 · Theorem proving (direct, contrapositive, contradiction, counterexample)

## Part A · Reading a statement precisely

### 1 STATEMENTS

Decide which of these are **statements** (they have a definite truth value) and which are **open sentences**. For each statement, say whether it is true or false and give the reason. For each open sentence, say what you would have to add to turn it into a statement.

- a.  $x + 1 = 4$ .
- b. For every real  $x$ ,  $x^2 \geq 0$ .
- c. There exists an integer  $n$  with  $n^2 = 9$ .
- d. There exists a real  $x$  with  $x^2 = -1$ .

**SOLUTION** (a) Open sentence: its truth depends on the unspecified  $x$  – true at  $x = 3$ , false at  $x = 0$ . Fixing a value of  $x$ , or putting a quantifier in front of it ("for every real  $x$ ...", "there exists a real  $x$ ..."), turns it into a statement.

(b) Statement, and **true**: the square of a real number is never negative.

(c) Statement, and **true**:  $n = 3$  is an integer and  $3^2 = 9$ . One witness is all an "there exists" claim needs. ( $n = -3$  works too, but one is enough.)

(d) Statement, and **false**: for every real  $x$  we have  $x^2 \geq 0 > -1$ , so no real number can satisfy  $x^2 = -1$ . Note what was required – a fact about *every* real  $x$ . Testing a few values would not have settled it.

## 2

## CONVERSE, CONTRAPOSITIVE

Over the domain  $\mathbb{Z}$ , consider: *if  $n$  is a multiple of 6, then  $n$  is even*.

- Write the converse and the contrapositive in words.
- Give the truth value of all three statements. Prove or disprove the original and the converse; for the contrapositive, you may quote a theorem from the lecture instead of proving it again.

**SOLUTION** (a) Converse: *if  $n$  is even, then  $n$  is a multiple of 6*. Contrapositive: *if  $n$  is not even, then  $n$  is not a multiple of 6* – for integers, "not even" is "odd".

(b) **Original: true.** Let  $n \in \mathbb{Z}$  be arbitrary and assume  $n = 6k$  for some  $k \in \mathbb{Z}$ . Then  $n = 2(3k)$  and  $3k \in \mathbb{Z}$ , so  $n$  is even.

**Converse: false.** Take  $n = 2$ : it is an integer and it is even, but  $2 = 6k$  has no integer solution, since  $k = \frac{1}{3} \notin \mathbb{Z}$ . So 2 is even and not a multiple of 6.

**Contrapositive: true**, and no new work is needed: a statement and its contrapositive are logically equivalent, so it is true because the original is.

## 3

## NEGATION

Write the negation of each statement in words, keeping the quantifiers in the right order. Then say which of the two – the statement or its negation – is true, and justify that one.

- For every real  $x$ ,  $x^2 > x$ .
- There exists an integer  $n$  with  $n^2 = 5$ .
- For every integer  $n$ , if  $n$  is prime then  $n$  is odd.

**SOLUTION** (a) Negation: *there exists a real  $x$  with  $x^2 \leq x$* . The negation is true:  $x = \frac{1}{2}$  is real and  $x^2 = \frac{1}{4} \leq \frac{1}{2}$ . (So the original is false.)

(b) Negation: *for every integer  $n$ ,  $n^2 \neq 5$* . The negation is true: if  $|n| \leq 2$  then  $n^2 \leq 4$ , and if  $|n| \geq 3$  then  $n^2 \geq 9$ ; no integer is left, so 5 is never reached. Note this needs an argument about every integer, not a few trials.

(c) Negation: *there exists an integer  $n$  such that  $n$  is prime and  $n$  is not odd*. The negation is true:  $n = 2$  is prime and even. Watch the shape – negating "if  $P$  then  $Q$ " gives " $P$  and not  $Q$ ", never "if  $P$  then not  $Q$ ".

#### 4 EMPTY HYPOTHESIS

An implication  $P \Rightarrow Q$  is false in one situation only:  $P$  true and  $Q$  false. So when **nothing satisfies the hypothesis  $P$** , no such situation can arise, there is no case left to check, and the implication is true automatically.

- Use this to explain, in two sentences, why "every integer  $n$  with  $n^2 < 0$  is prime" is true.
- Write a statement of your own about real numbers that is true for the same reason – because no real number satisfies its hypothesis – and say in one sentence why it tells the reader nothing.

**SOLUTION** (a) An implication is false only when its hypothesis holds and its conclusion fails. No integer satisfies  $n^2 < 0$ , since the square of an integer is never negative, so that situation never arises and the statement is true.  
(b) For example: *every real  $x$  with  $x^2 + 1 = 0$  satisfies  $x = 7$* . It is true because no real number satisfies the hypothesis, and it is useless for exactly the same reason: it describes no actual number, so nothing can be deduced from it. A proof that works only because its hypothesis is never satisfied has proved nothing worth having.

## Part B · Writing the proof

Name the method at the top of each solution – direct, contrapositive, contradiction or counterexample – then write the proof in full sentences: state the domain, take an **arbitrary** element of it, and finish with the definition the conclusion asks for.

#### 5 DIRECT PROOF

Prove: for all integers  $m$  and  $n$ , if  $m$  and  $n$  are odd, then  $m + n$  is even.

**SOLUTION** Method: direct proof. Let  $m, n \in \mathbb{Z}$  be arbitrary and assume both are odd. By the definition of odd there are integers  $r$  and  $s$  with

$$m = 2r + 1 \quad \text{and} \quad n = 2s + 1.$$

Adding,

$$m + n = (2r + 1) + (2s + 1) = 2r + 2s + 2 = 2(r + s + 1).$$

Since  $r, s \in \mathbb{Z}$ , also  $r + s + 1 \in \mathbb{Z}$ , so  $m + n$  is 2 times an integer – which is the definition of even. As  $m$  and  $n$  were arbitrary, this holds for all odd integers. ■

Each claim below is either true or false. Decide which, then prove it or disprove it with one counterexample. For a counterexample, name the value, confirm it lies in the stated domain, and show exactly what fails.

- For every real  $x$ : if  $x^2 > x$ , then  $x > 1$ .
- For every integer  $n$ ,  $n^2 + n$  is even.
- For all real  $a$  and  $b$ :  $|a + b| = |a| + |b|$ .

**SOLUTION** (a) **False** – counterexample. Take  $x = -2$ , which is real. The hypothesis holds, since  $x^2 = 4 > -2 = x$ ; the conclusion fails, since  $-2 > 1$  is false. One case with a true hypothesis and a false conclusion is exactly what refutes an implication.

(b) **True** – direct proof. Let  $n \in \mathbb{Z}$  be arbitrary. Then  $n^2 + n = n(n + 1)$ , a product of two consecutive integers. One of  $n$  and  $n + 1$  is even: if  $n$  is even we are done, and if  $n$  is odd then  $n = 2k + 1$ , so  $n + 1 = 2(k + 1)$  is even. Writing that even factor as  $2j$  with  $j \in \mathbb{Z}$  gives  $n^2 + n = 2j \cdot (\text{the other factor})$ , which is 2 times an integer, hence even. ■

(c) **False** – counterexample. Take  $a = 1$  and  $b = -1$ , both real. Then  $|a + b| = |0| = 0$ , while  $|a| + |b| = 1 + 1 = 2$ , and  $0 \neq 2$ . (The true statement is the triangle inequality  $|a + b| \leq |a| + |b|$ , with equality only when  $a$  and  $b$  have the same sign or one of them is 0.)

Prove: for every integer  $n$ , if  $3n + 2$  is even, then  $n$  is even. Say in one sentence why starting from the hypothesis directly is awkward here.

**SOLUTION** **Method: contrapositive.** For integers, "not even" means "odd", so the contrapositive reads: *if  $n$  is odd, then  $3n + 2$  is odd.*

Let  $n \in \mathbb{Z}$  be arbitrary and assume  $n$  is odd, so  $n = 2k + 1$  for some  $k \in \mathbb{Z}$ . Then

$$3n + 2 = 3(2k + 1) + 2 = 6k + 5 = 2(3k + 2) + 1.$$

Since  $3k + 2 \in \mathbb{Z}$ , the number  $3n + 2$  has the form  $2 \cdot (\text{integer}) + 1$ , so it is odd. This proves the contrapositive, and a statement and its contrapositive are logically equivalent, so the original claim holds. ■

Why not directly: the hypothesis " $3n + 2$  is even" gives  $3n + 2 = 2m$ , i.e.  $3n = 2(m - 1)$ , and concluding that  $n$  itself is even from that needs the extra fact that 3 is odd.

Assuming  $n$  is odd hands you  $n = 2k + 1$  immediately, which is algebra you can use at once.

## 8 DIRECT · DIVISIBILITY

Prove: for every odd integer  $n$ , the number  $n^2 - 1$  is divisible by 8. (You may use, with a one-line justification, that the product of two consecutive integers is even.)

**SOLUTION** Method: direct proof. Let  $n$  be an arbitrary odd integer, so  $n = 2k + 1$  for some  $k \in \mathbb{Z}$ . Then

$$n^2 - 1 = (2k + 1)^2 - 1 = 4k^2 + 4k + 1 - 1 = 4k(k + 1).$$

Now  $k(k + 1)$  is a product of two consecutive integers, so one of them is even and the product is even: write  $k(k + 1) = 2m$  with  $m \in \mathbb{Z}$ . Therefore

$$n^2 - 1 = 4 \cdot 2m = 8m,$$

with  $m \in \mathbb{Z}$ , which is exactly what "divisible by 8" means. ■

The step worth noticing:  $4k(k + 1)$  alone only gives divisibility by 4. The factor 8 appears only after using the parity of  $k(k + 1)$  – dropping that line loses the claim.

## 9 CONTRADICTION

Prove: if  $a$  is rational and  $b$  is irrational, then  $a + b$  is irrational.

**SOLUTION** Method: contradiction. Assume  $a$  is rational,  $b$  is irrational, and – for contradiction – that  $a + b$  is rational. By the definition of rational there are integers  $p, q, r, s$  with  $q \neq 0, s \neq 0$  and

$$a = \frac{p}{q}, \quad a + b = \frac{r}{s}.$$

Subtracting  $a$  from both sides of the second equation,

$$b = \frac{r}{s} - \frac{p}{q} = \frac{rq - ps}{sq}.$$

Here  $rq - ps \in \mathbb{Z}$  and  $sq \in \mathbb{Z}$ , and  $sq \neq 0$  because neither  $s$  nor  $q$  is 0. So  $b$  is a ratio of two integers with non-zero denominator, i.e.  $b$  is rational. That contradicts the assumption that  $b$  is irrational. Hence  $a + b$  cannot be rational, so it is irrational. ■

The contradiction is a genuine consequence of the assumption – " $b$  is rational" against " $b$  is irrational" – not merely a surprising outcome.

10 UNIQUENESS

Prove that there is **at most one** real number  $e$  with the property that  $x + e = x$  for every real  $x$ . Then name a number with the property, and say what the two facts together establish.

**SOLUTION** Method: the uniqueness shape – assume two objects qualify, show they are equal.

Suppose  $e$  and  $e'$  are both real numbers with the property, that is,

$$x + e = x \text{ for every real } x, \quad x + e' = x \text{ for every real } x.$$

Apply the first with  $x = e'$ :  $e' + e = e'$ . Apply the second with  $x = e$ :  $e + e' = e$ .

Addition of real numbers is commutative, so  $e' + e = e + e'$ , and therefore

$$e' = e' + e = e + e' = e.$$

So any two numbers with the property are equal: there is at most one. ■

The number  $e = 0$  has the property, since  $x + 0 = x$  for every real  $x$ . "At most one" plus "at least one" gives **exactly one** – 0 is the unique additive identity. Note that the uniqueness argument never needed to know which number it was.

## Part C · Reading and writing proofs

11 FIND THE ERROR

A student hands in the following.

**Claim.** For every  $n \in \mathbb{Z}$ , if  $n^2$  is even, then  $n$  is even.

**Proof.** Assume  $n$  is even. Then  $n = 2k$  for some  $k \in \mathbb{Z}$ , so  $n^2 = (2k)^2 = 2(2k^2)$  with  $2k^2 \in \mathbb{Z}$ , so  $n^2$  is even, which proves the claim. ■

- Every line of algebra here is correct. Which statement has actually been proved, and what is the name of the mistake?
- Give a correct proof of the claim.

**SOLUTION** (a) The algebra proves *if  $n$  is even, then  $n^2$  is even* – the **converse** of the claim. The student assumed the conclusion and derived the hypothesis, which is the wrong direction; a converse is a different statement and can have a different truth value. (Here the converse happens to be true as well, which is what makes the error easy to miss: a correct proof of a true statement that is not the one asked for.)

(b) **Method: contrapositive.** The contrapositive is: *if  $n$  is odd, then  $n^2$  is odd*. Let  $n \in \mathbb{Z}$  be arbitrary and odd, so  $n = 2k + 1$  with  $k \in \mathbb{Z}$ . Then

$$n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1,$$

and  $2k^2 + 2k \in \mathbb{Z}$ , so  $n^2$  is odd. The contrapositive holds, hence so does the claim. ■

In three or four sentences, explain why proving the contrapositive proves the original statement, while proving the converse does not. Include one concrete example of your own.

**WHAT A GOOD ANSWER DOES**

A strong answer says *when* an implication fails:  $P \Rightarrow Q$  is false in exactly one situation, namely  $P$  true and  $Q$  false – and  $\neg Q \Rightarrow \neg P$  fails in exactly that same situation, so the two are true in exactly the same cases. The converse  $Q \Rightarrow P$  fails in a different situation ( $Q$  true,  $P$  false), so its truth value is unrelated.

The example should show a true statement whose converse is false – for instance "multiple of 6  $\Rightarrow$  even" is true, its converse fails at  $n = 2$ , and its contrapositive "odd  $\Rightarrow$  not a multiple of 6" is true along with the original. Answers that only restate the definitions without saying *when each form fails* have described the shapes without explaining the equivalence.